

EE 705: VLSI Design Lab

Course Project-01 Report



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1 Problem Statement

The objective of this project is to design and implement an SPI-Lite module for a RISC-V processor using the AXI-Lite interface. The module facilitates SPI communication while ensuring compatibility with AXI-Lite protocol standards. .

1.1 Functional Description

The SPI-Lite module follows the SPI protocol and integrates with the AXI-Lite interface. The key functions include:

- Address decoding for register access.
- Configuration and status handling using registers such as:
 - **SPI_DGIER** (Global Interrupt Enable Register)
 - **SPI_IPISR** (Interrupt Status Register)
 - **SPI_IPIER** (Interrupt Enable Register)
 - **SPI_SR** (Status Register)
 - **SPI_CR** (Control Register)
 - **SPI_DTR** (Data Transmit Register)
 - **SPI_DRR** (Data Receive Register)
- AXI-Lite compliant read and write transactions.
- SPI transaction control, including FIFO buffering.
- Generating interrupts for events like TX FIFO empty or RX FIFO full.

1.2 Implementation and Testing

- The SPI-Lite module is implemented using 5 different blocks.
- A testbench is created for wrapper.
- The integration ensures compliance with the RISC-V ISA (RV32I).

2 Block Diagram of Entire SPI-Lite Module

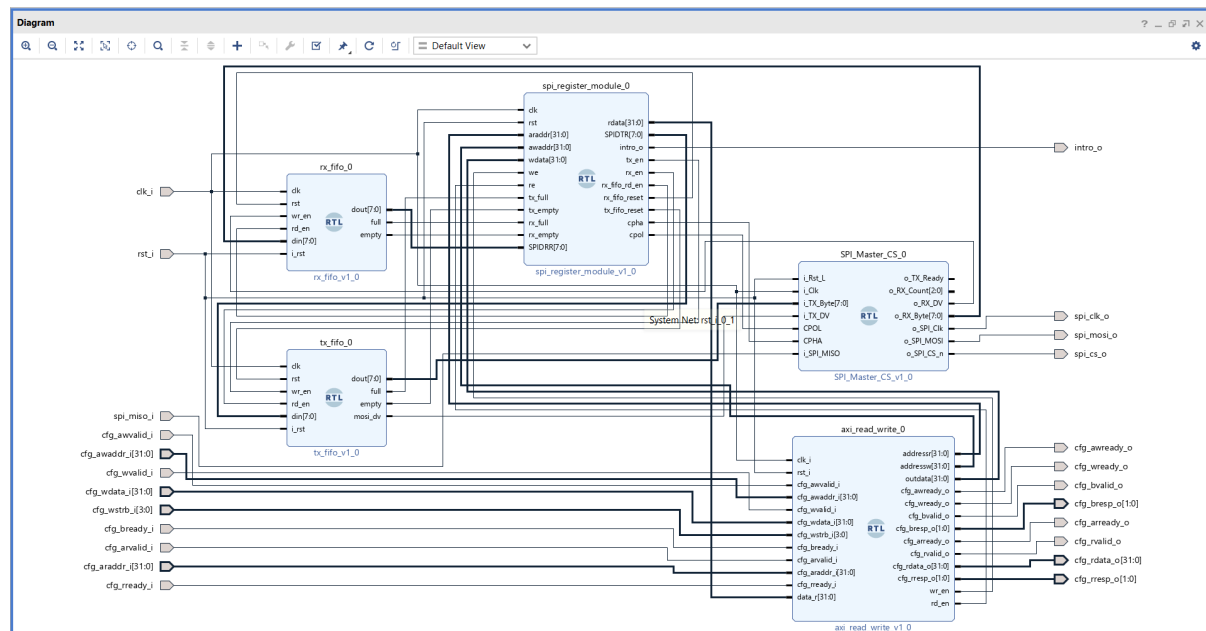


Figure 1: Block Diagram

3 Utilization Report of SPI-Lite Module

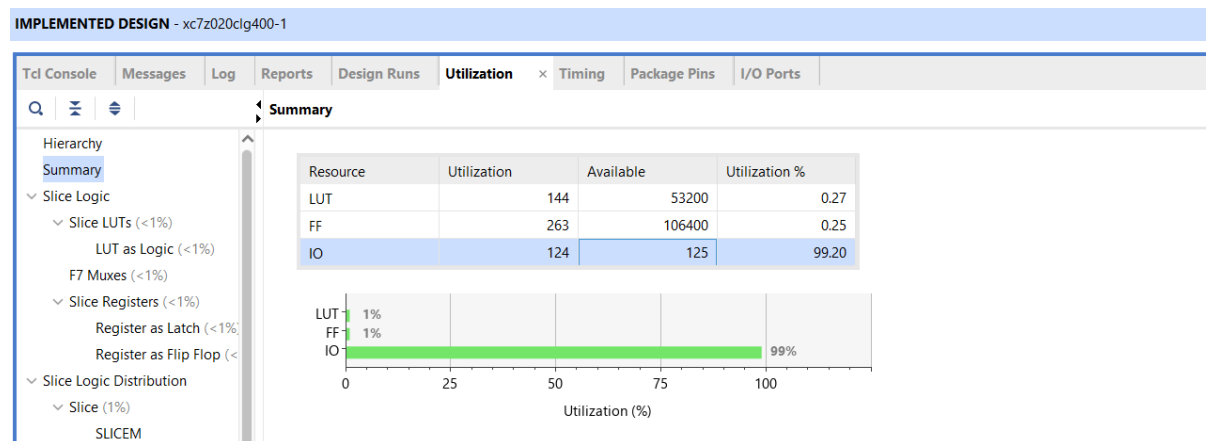


Figure 2: Utilization Report

4 Simulations

4.1 Behavioral Simulation of spi lite module

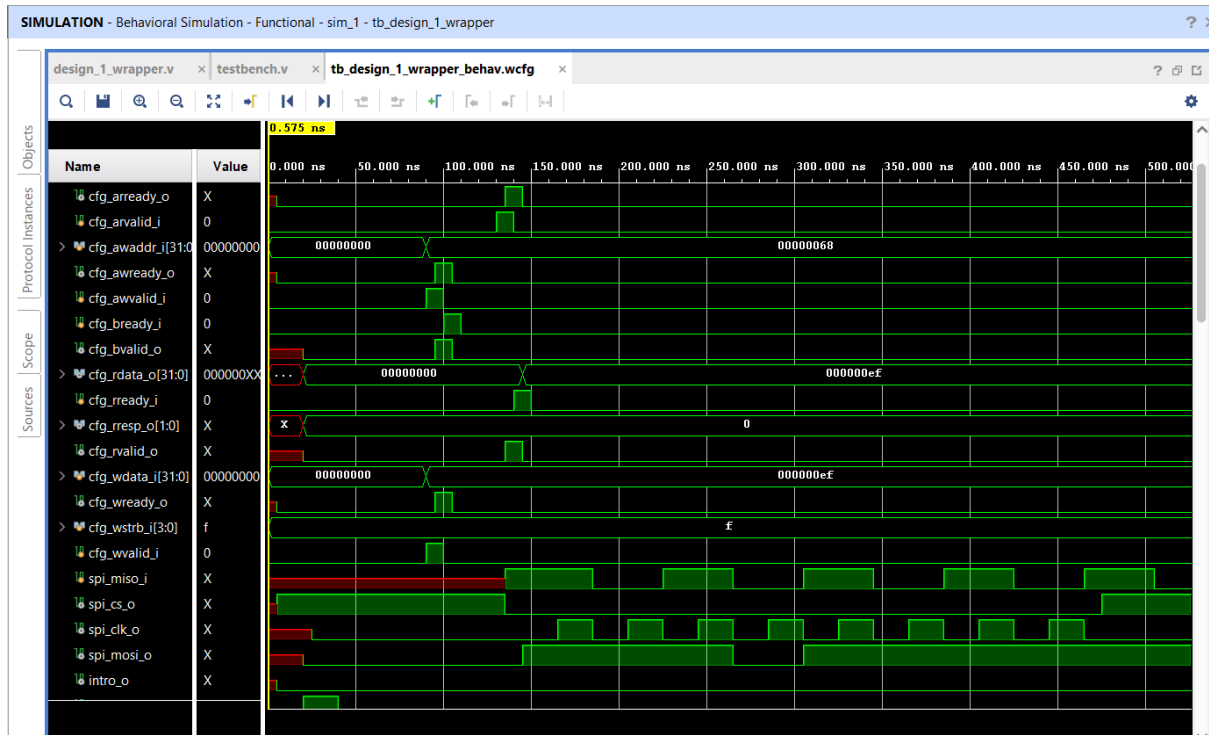


Figure 3: Behavioral Simulation of spi lite module

SIMULATION - Post-Synthesis Analysis - Timing - sim_1 - tb_design_1_wrapper

design_1_wrapper.v x **tb_design_1_wrapper_behav.wcfg** x testbench.v x

0.575 ns

Name	Value
cfg_arready_o	X
cfg_arvalid_i	0
cfg_awaddr_i[31:0]	00000000
cfg_awaitready_o	X
cfg_awaitvalid_i	0
cfg_bready_i	0
cfg_bvalid_o	X
cfg_rdata_o[31:0]	XXXXXXXX
cfg_rready_i	0
cfg_rresp_o[1:0]	X
cfg_rvalid_o	X
cfg_wdata_i[31:0]	00000000
cfg_wready_o	X
cfg_wstrb_i[3:0]	f
cfg_wvalid_i	0
spi_miso_i	X
spi_cs_o	X
spi_clk_o	X
spi_mosi_o	X
intro_o	X

4

4.3 Post Implementation Simulation

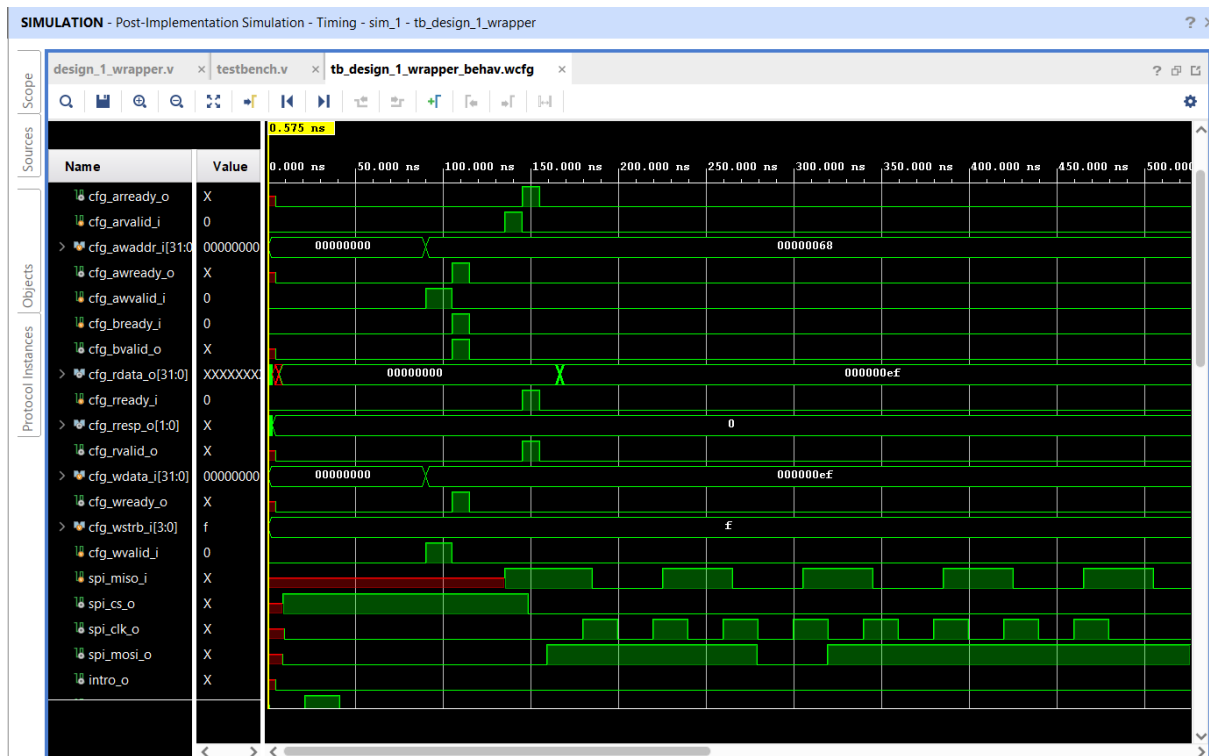


Figure 5: Post Implementation Simulation of spi lite module

5 Individual Blocks Verification

5.1 AXI Block

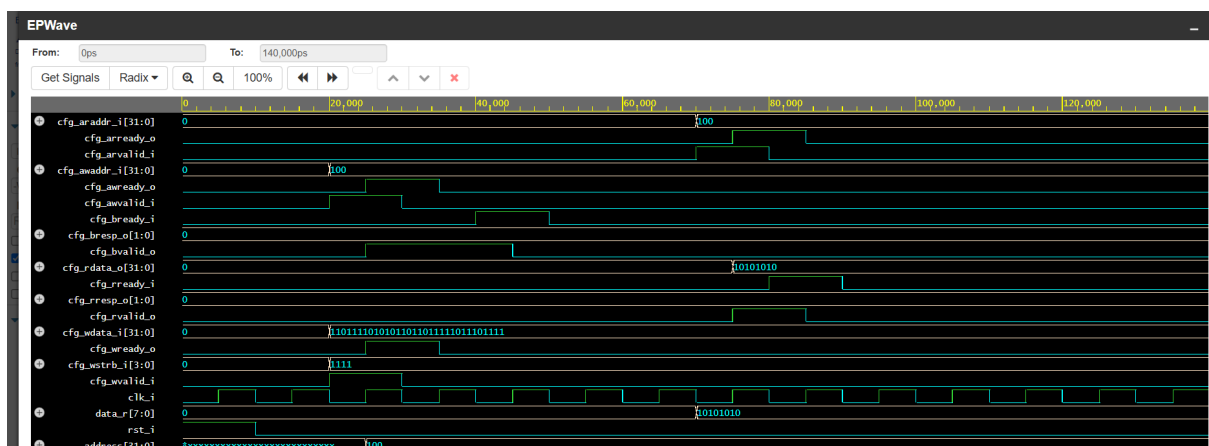


Figure 6: Simulation waveforms of AXI Block

5.5 SPI Module

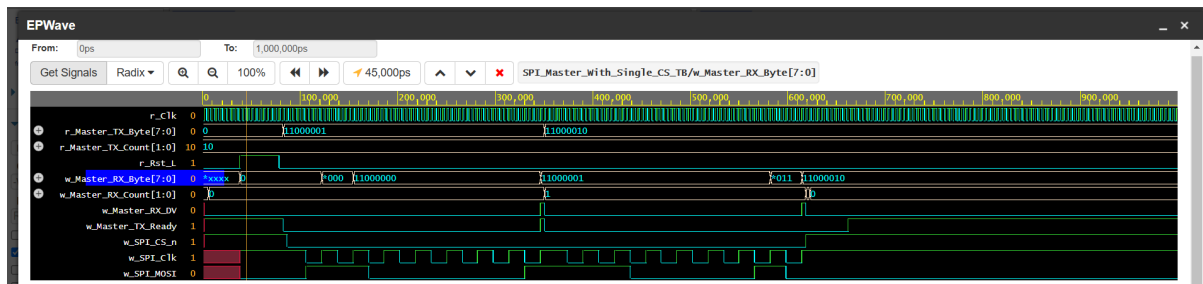


Figure 10: Simulation waveforms of SPI Module

6 Interface Diagram of how SPI lite interacts with other module(s)

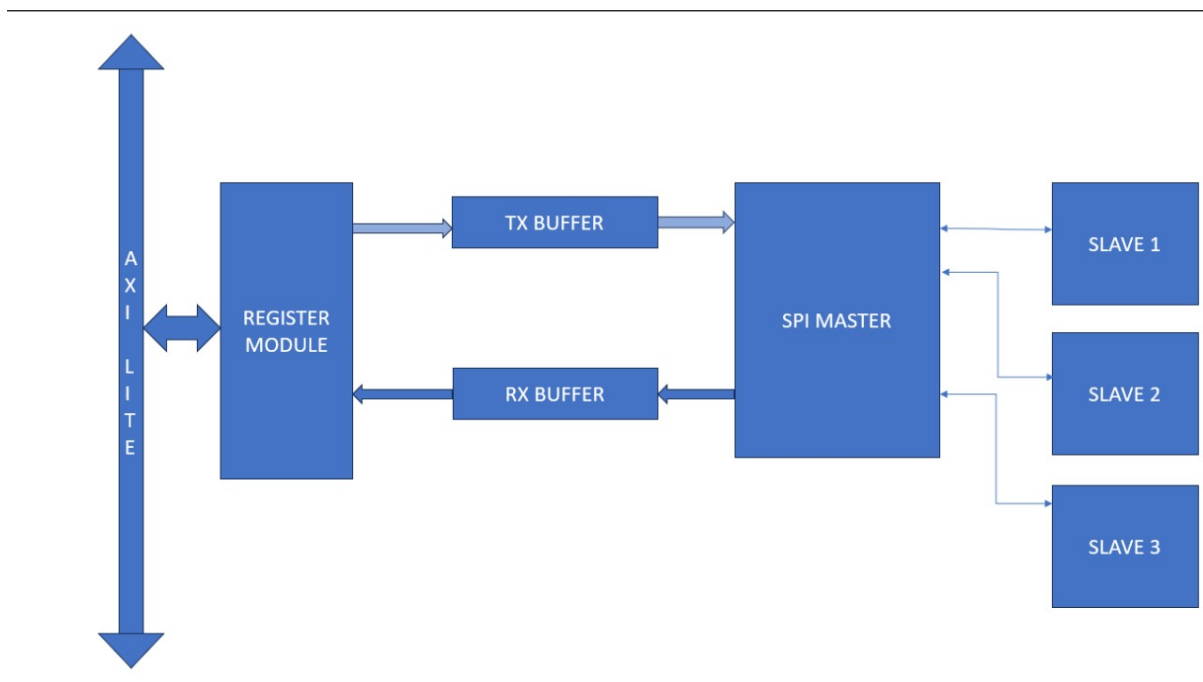


Figure 11: Interface Diagram

Our SPI lite module can only interface with the AXI Lite signals and can communicate between AXI Lite and other SPI Slave devices

NOTE:-INTERRUPTs ARE ADDED IN THE REGISTER MODULE ITSELF

7 References

- <https://docs.amd.com/internal/api/webapp/print/f5d7e1fe-93c9-4170->

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